M3503 WEEK 2 2021

6. BERNOULLI EQUATION

It is an equation with the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$, where n is a real number other than 1 or 0.

N.B If $n=1$ or $n=0$, we have a linear equation

To solve Bernoulli equation, we make the substitution $u = y^{1-n}$, which transforms the equation into a linear equation.

Examples

Solve ① $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$

② $x\frac{dy}{dx} + y = \frac{1}{y^2}$

③ $\frac{dy}{dx} - \frac{y}{x} = \sqrt{\frac{y}{x}}$

Sol^n ① $\frac{dy}{dx} + \frac{y}{x} = x^2y^2 \Rightarrow y^{-2}\frac{dy}{dx} + \frac{y^{-1}}{x} = x^2$

let $u = y^{-1} \Rightarrow \frac{du}{dx} = -y^{-2}\frac{dy}{dx}$

Substituting, we get

$$-\frac{du}{dx} + \frac{u}{x} = x^2 \Rightarrow \frac{du}{dx} - \frac{u}{x} = -x^2$$

Integrating factor $= \exp\left(\int -\frac{1}{x}\,dx\right)$

$= \exp(-\ln x)$

$= \frac{1}{x}$

So $\frac{d}{dx}\left(\frac{y}{x}\right) = -x$

$\Rightarrow \frac{y}{x} = -\frac{1}{2}x^2 + C$

$\Rightarrow y^{-1}x^{-1} = -\frac{1}{2}x^2 + C$ or $y = 0$ ($y=0$ got lost as a sol^{n} since we divided by y)

2. $x\frac{dy}{dx} + y = \frac{1}{y^2} \Rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{1}{xy^2}$

$\Rightarrow y^2\frac{dy}{dx} + \frac{y^3}{x} = \frac{1}{x}$

Let $u = y^3$

$\Rightarrow \frac{du}{dx} = 3y^2\frac{dy}{dx}$

Substitution gives

$\frac{1}{3}\frac{du}{dx} + \frac{u}{x} = \frac{1}{x}$

Integrating factor $= \exp\left(\int \frac{3}{x}dx\right)$

$= \exp(3\ln x)$

$= x^3$

So, $\frac{d}{dx}(ux^3) = 3x^2$

$\Rightarrow ux^3 = x^3 + C$

$\Rightarrow (xy)^3 = x^3 + C$

3. $\frac{dy}{dx} - \frac{y}{x} = \sqrt{\frac{y}{x}} \Rightarrow y^{-1/2}\frac{dy}{dx} - \frac{y^{1/2}}{x} = \frac{1}{\sqrt{x}}$

Let $u = y^{1/2}$

$\Rightarrow \frac{du}{dx} = \frac{1}{2} y^{-1/2} \frac{dy}{dx}$

$\Rightarrow 2\frac{du}{dx} - \frac{u}{x} = \frac{1}{\sqrt{x}}$.

Integrating factor $= \exp\left(\int -\frac{1}{2x}\, dx\right)$

$= \exp\left(-\frac{1}{2}\ln x\right)$

$= x^{-1/2}$

$\therefore \frac{d}{dx}(ux^{-1/2}) = \frac{1}{2x}$.

$\Rightarrow ux^{-1/2} = \frac{1}{2}\ln x + C$

$\Rightarrow \sqrt{\frac{y}{x}} = \ln(\sqrt{x}) + C$

7. EQUATIONS WITH LINEAR COEFFICIENTS

These are equations with the form

$$(a_1x + b_1y + c_1)\,dx + (a_2x + b_2y + c_2)\,dy = 0 \quad \text{— (1)}$$

When $\frac{a_1}{b_1} = \frac{a_2}{b_2}$, eqn (1) becomes

$$\frac{dy}{dx} = -\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$$

$$= -\frac{b_1\left(\frac{a_1}{b_1}x + y\right) + c_1}{b_2\left(\frac{a_1}{b_1}x + y\right) + c_2}$$

which is of the form $\frac{dy}{dx} = F(ax + by)$

- When $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$ and $c_1 = c_2 = 0$, eq^n ① becomes

$$\frac{dy}{dx} = -\frac{a_1x + b_1y}{a_2x + b_2y}$$

which is homogeneous

- Generally, when $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$, we make the substitutions $x = u + k$ and $y = v + c$, where k and c are constants satisfying the equations

$$a_1k + b_1c + c_1 = 0$$
$$a_2k + b_2c + c_2 = 0$$

These substitutions transform eq^n ① into

$$\frac{dv}{du} = -\frac{a_1u + b_1v}{a_2u + b_2v}$$ which is homogeneous

Examples

Solve ① $(-3x + y - 1)dx + (x + y + 3)dy = 0$

② $(x + 2y - 4)dx - (2x - 4y)dy = 0$

③. $(x - 2y)dx + (2x + y + 5)dy = 0$

soln

①. $\frac{dy}{dx} = \frac{3x - y + 1}{x + y + 3}$

Auxiliary eq^{n}s: $3k - c + 1 = 0$, $k + c + 3 = 0$ $\Rightarrow 4k + 4 = 0$

$\Rightarrow k = -1$

$\Rightarrow c = -2$

Let $x = u - 1$ and $y = v - 2$.

$\Rightarrow dx = du$, $dy = dv$

So, $\frac{dv}{du} = \frac{3u - 3 - v + 2 + 1}{u - 1 + v - 2 + 3} = \frac{3u - v}{u + v}$ which is homogeneous

Let $V = tu \Rightarrow \frac{dv}{du} = t + u\frac{dt}{du}$

So, $t + u\frac{dt}{du} = \frac{3u - tu}{u + tu} = \frac{3-t}{1+t}$

$\Rightarrow u\frac{dt}{du} = \frac{3 - t - t - t^2}{1+t}$

$\Rightarrow \frac{1+t}{(t+3)(t-1)}\,dt = -\frac{du}{u}$

$\Rightarrow \left[\frac{1}{2(t+3)} + \frac{1}{2(t-1)}\right] dt = -\frac{du}{u}$

$\Rightarrow \frac{1}{2}\ln(t+3) + \frac{1}{2}\ln(t-1) = -\ln(u) + \ln C$

$\Rightarrow \ln[(t+3)(t-1)u^2] = \ln C$

$\Rightarrow (t+3)(t-1)u^2 = C$

$\Rightarrow \left(\frac{v+3u}{u}\right)\left(\frac{v-u}{u}\right)u^2 = C$

$\Rightarrow (y+2+3x+3)(y+2-x-1) = C$

$\Rightarrow (y+3x+5)(y-x+1) = C$

(2) $\frac{dy}{dx} = \frac{x+2y-4}{2x-4y}$

Auxiliary eqn: $\left.\begin{matrix} k+2c-4=0 \\ 2k-4c=0 \end{matrix}\right\} \Rightarrow 4k-8=0$

$\Rightarrow k=2$

$\Rightarrow c=1$

Let $x = u+2$, $y = v+1$

$\Rightarrow dx = du$, $dy = dv$

So, $\frac{dv}{du} = \frac{u+2+2v+2-4}{2u+4-4v-4}$

$= \frac{u+2v}{2u-4v}$, which is homogeneous

Let $v = tu \Rightarrow \frac{dv}{du} = t + u\frac{dt}{du}$

then, $t + u\frac{dt}{du} = \frac{-(u+2tu)}{4tu-2u}$

$\Rightarrow u\frac{dt}{du} = \frac{2t-4t^2-2t-1}{4t-2}$

$= -\frac{[4t^2+1]}{2(2t-1)}$

$\Rightarrow \frac{2t-1}{4t^2+1}\,dt = -\frac{1}{2u}\,du$

$\Rightarrow \left[\frac{2t}{4t^2+1} - \frac{1}{4t^2+1}\right]dt = \frac{-1}{2u}\,du$

$\Rightarrow \frac{1}{4}\ln|4t^2+1| + \frac{1}{2}\tan^{-1}(2t) = -\frac{1}{2}\ln u + \frac{1}{2}\ln c$

$\Rightarrow \frac{1}{4}\ln\left|4\left(\frac{y}{u}\right)^2+1\right| + \frac{1}{2}\tan^{-1}\left(\frac{2v}{u}\right) = \frac{1}{2}\ln c - \frac{1}{2}\ln u$

$\Rightarrow \frac{1}{2}\ln\left|4\left[\frac{y-1}{x-2}\right]^2+1\right| + \tan^{-1}\left(\frac{2y-2}{x-2}\right) = \ln c - \ln|x-2|$

② $\frac{dy}{dx} = \frac{2y-x}{2x+y+5}$

Auxiliary eqns: $2C - K = 0$, $2K + C + 5 = 0$ $\Rightarrow 5C + 5 = 0$

$\Rightarrow C = -1$

$\Rightarrow K = -2$

let $x = u - 2$, $y = v - 1$

So that $\frac{dv}{du} = \frac{2v - 2 - u + 2}{2u - 4 + v - 1 + 5}$

$= \frac{2v - u}{2u + v}$ *

let $v = tu \Rightarrow \frac{dv}{du} = t + u\frac{dt}{du}$

So, $t + u\frac{dt}{du} = \frac{2tu - u}{2u + tu}$

$\Rightarrow u\frac{dt}{du} = \frac{2tu - u - 2tu - t^2u}{2u + tu}$

$= -\frac{(t^2+1)}{t+2}$

$\Rightarrow \frac{t+2}{t^2+1}\,dt = -\frac{1}{u}\,du$

$\Rightarrow \frac{t\,dt}{t^2+1} + \frac{2\,dt}{t^2+1} = -\frac{1}{u}\,du$

$\Rightarrow \frac{1}{2}\ln|t^2+1| + 2\tan^{-1}(t) = -\ln u + \ln C$

$\Rightarrow \frac{1}{2}\ln\left|\left(\frac{v}{u}\right)^2+1\right| + 2\tan^{-1}\left(\frac{v}{u}\right) = -\ln u + \ln C$

$\Rightarrow \frac{1}{2}\ln\left|\left(\frac{y+1}{x+2}\right)^2+1\right| + 2\tan^{-1}\left(\frac{y+1}{x+2}\right) = -\ln|x+2| + \ln C$

8. RICCATI EQUATION

It is an equation of the form

$$\frac{dy}{dx} = A(x)y^2 + B(x)y + C(x) \quad \text{—①}$$

where $A(x)$, $B(x)$, $C(x)$ are continuous functions of x.

The general solution of eq^n ① is given by

$$y = y_1 + u$$

where y_1 is a particular solution of eq^n ①.

N.B The general solution can be found if a particular solution is known.

Examples

solve ① $x^3y' + x^2y - y^2 = 2x^4$; $y_1 = x^2$

② $xy' - 2y = y^2 - x^4$; $y_1 = x^2$

③ $y' + 2xy = 1 + x^2 + y^2$; $y_1 = x$

Sol^n

① $y = y_1 + u = x^2 + u$

$\Rightarrow y' = 2x + u'$

Substitution gives

$$x^3(2x+u') + x^2(x^2+u) - (x^2+u)^2 = 2x^4$$

$$\Rightarrow x^3u' + 2x^4 + x^4 + x^2u - x^4 - 2x^2u - u^2 = 2x^4$$

$$\Rightarrow x^3u' - x^2u - u^2 = 0$$

$$\Rightarrow u' - x^{-1}u - x^{-3}u^2 = 0$$ which is Bernoulli

$$\Rightarrow u^{-2}u' - x^{-1}u^{-1} - x^{-3} = 0$$

Let $v = -u^{-1} \Rightarrow v' = u^{-2}u'$

So that, $v' + vx^{-1} - x^{-3} = 0$, which is linear

Integrating factor $= \exp\left(\int \frac{1}{x}\,dx\right) = x$

Multiplying the linear equation by the integrating factor gives

$$xv' + v = x^{-2}$$
$$\Rightarrow \frac{d(vx)}{dx} = x^{-2}$$
$$\Rightarrow vx = -\frac{1}{x} + C$$
$$\Rightarrow v = -\frac{1}{x^2} + \frac{C}{x}$$
$$\Rightarrow -\frac{1}{u} = -\frac{1}{x^2} + \frac{C}{x}$$
$$\Rightarrow \Rightarrow \frac{1}{u} = \frac{-Cx + 1}{x^2}$$
$$\Rightarrow u = \frac{x^2}{1 - Cx}$$

$$\therefore y = x^2 + \frac{x^2}{1 - Cx}$$

② Let $y = y_1 + u = x^2 + u$
$\Rightarrow y' = 2x + u'$

Upon substituting, we get
$$2x^2 + xu' - 2x^2 - 2u = x^4 + 2x^2u + u^2 - x^4$$
$$\Rightarrow xu' - 2u - 2x^2u - u^2 = 0$$
$$\Rightarrow u' - 2(x^{-1} + x)u - x^{-1}u^2 = 0$$, which is Bernoulli

let $v = -u^{-1} \Rightarrow v' = u^{-2}u'$

Substitution gives

$$v' + v(2x^{-1} + 2x) - x^{-1} = 0 \quad \text{which is linear}$$

Integrating factor $= \exp\left(\int \left(\frac{2}{x} + 2x\right) dx\right)$

$$= e^{(\ln x^2 + x^2)}$$
$$= x^2 e^{x^2}$$

$$x^2 e^{x^2} v' + v(2x^{-1} + 2x) x^2 e^{x^2} = x e^{x^2}$$
$$\Rightarrow x^2 e^{x^2} v = \tfrac{1}{2} e^{x^2} + C$$
$$\Rightarrow v = \frac{1}{2x^2} + \frac{C}{x^2 e^{x^2}}$$

(5) $y' + 2xy = 1 + x^2 + y^2$, $y_1 = x$

let $y = x + u$

$$\Rightarrow y' = 1 + u'$$

Upon substituting, we get

$$1 + u' + 2x(x+u) = 1 + x^2 + (x+u)^2$$
$$\Rightarrow 1 + u' + 2x^2 + 2xu = 1 + x^2 + x^2 + 2xu + u^2$$
$$\Rightarrow u' = u^2$$
$$\Rightarrow u^{-2} du = dx$$
$$\Rightarrow -\frac{1}{u} = x + C$$
$$\Rightarrow -\frac{1}{y - x} = x + C$$
$$\Rightarrow y = -\frac{1}{x + C} + x.$$